



CLEARANCE

Active TS Clearance

TECHNICAL SKILLS

- Programming: C++, Python, Java.
- Software Engineering: Object-Oriented Design (OOD), Agile/Scrum.
- DevSecOps: GitLab CI/CD, Jenkins, Kubernetes, Docker.
- System & Cloud: Linux, AWS, Red Hat Enterprise Linux (RHEL).
- Testing & Validation: Unit Testing, Integration Testing, and Debugging.
- Modeling & Simulation: 6-DOF Flight Dynamics, Missile Simulation and Monte Carlo Analysis.

PROFESSIONAL EXPERIENCE

Software Design Engineer III, Torch Technologies, Feb 2026 - Current

- Developed C++ and Python features within model development and simulation execution frameworks, integrating updates across three software components and streamlining debugging workflows used for model validation.
- Diagnosed and resolved complex C++ and Python software defects using Linux, Visual Studio, Git, and automated unit and integration testing, reducing recurring issues by approximately 10% and improving software stability.
- Translated system requirements into software designs, code implementations, and verification documentation, reducing integration rework by approximately 15% and improving delivery efficiency.
- Collaborated with software, systems, and simulation engineers to design, integrate, and validate software enhancements across shared simulation baselines, ensuring compatibility and reliable end-to-end functionality.
- Designed, implemented, and validated high-fidelity C++ six-degree-of-freedom (6-DOF) missile simulation models, including autopilot control logic and the integration of guidance, sensor, and flight-dynamics components for LRPM and Hydra-70, increasing simulation throughput by 10%.

Software Engineer II, BAE Systems , Sep 2024 - Jan 2026, Utah

- Developed, reviewed, and tested C++11 six-degree-of-freedom (6-DOF) flight-dynamics models supporting LGM-35A Sentinel ICBM boost and post-boost performance, maintaining 100% traceability between software implementation and flight-performance requirements.
- Diagnosed and resolved simulation inconsistencies by analyzing flight behavior, debugging C++ model logic, and refining model assumptions, improving model fidelity and reducing downstream verification and validation effort.
- Implemented and validated software updates through code reviews, regression testing, and requirements-based analysis, helping maintain reliable simulation performance across evolving model baselines.

Software Engineer I, Space Dynamics Laboratory, Jan 2023 - Sep 2024, Utah

- Developed and maintained C++ and Python software modules within the FORGE MDPAF ground framework, implementing enhancements, debugging issues, and supporting reliable data processing capabilities.
- Collaborated with cross-functional engineering teams to analyze requirements, troubleshoot software defects, and deliver high-quality solutions that improved system reliability and performance.

Information Technology Specialist, Air Force, Jan 2019 - Dec 2022

- Provided technical support for 100+ users, troubleshooting hardware, software, and network issues.
- Installed, configured, and maintained 100+ workstations help ensuring 98% system availability and reliable support for daily military operations.

PROJECTS

- Mission Planning Dashboard: Built a full-stack mission-planning dashboard with role-based workflows, REST API endpoints, and audit logging, reducing planning data-entry errors by 30%.
- Missile Flight Dynamics Simulation: Built a standalone C++ 6-DOF missile simulation with autopilot control logic, guidance interfaces, and flight-dynamics behavior; validated model performance through Monte Carlo analysis.

EDUCATION

Master of Science, Aerospace Engineering Colorado State University	Dec 2033
Master of Science, Computer Science Georgia Institute of Technology	May 2027
Bachelor of Science, Computer Science Valdosta State University	Dec 2022

CERTIFICATIONS

CISSP | CKA | CKS | AWS Solutions Architect | Security+ | PMP

PUBLICATION

- Co-authored peer-reviewed Springer publication on distributed fault-containment simulation using randomized scheduling, published in *Advanced Communication and Intelligent Systems*.